

## Lee Ming Xiang

Applied Scientist | Subsurface Domain Data Scientist | Geophysicist

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### About Me

Subsurface Applied Scientist with Petroleum Geoscience & Geophysics background, 8+ years of experience across Oil & Gas & AI industry, with technical leadership with end-to-end development and deployment of AI solutions. Proven expertise in **LLM domain adaptation post training**, **Multimodal AI**, designed **Multi-Lingual & Multi-Agent LLM systems**, **Multimodal Retrieval-Augmented Generation (RAG)**, and **computer vision** for Seismic AI solution development. Experienced in **leading and mentoring data scientists and engineering team**, while remain hands-on across model training, evaluation and product deployment and integration. Collaborates with **cross-functional and cross-domains teams**, **AI scientists**, and engaging with **global clients** to drive **AI adoption for subsurface** domain workflows.

### Education

**University Malaya** - Faculty of Computer Science & Information Technology Malaysia  
**Master of Data Science – Graduated with Distinction** 2020-2022  
Thesis Title: [Carbon Dioxide Emission Monitoring based on Prediction of Gas Fuel Rate using Machine Learning](#)

**University of Technology Petronas** – Faculty of Geoscience and Petroleum Engineering Malaysia  
**Bachelor of Science (Hons) in Petroleum Geoscience – Gold Vice Chancellor Award** 2012-2017  
Major: Exploration Geophysics  
Thesis Title: [Application of Simultaneous Inversion Characterizing Reservoir Properties in X Field, Sabah Basin](#)

**Missouri University of Science & Technology** – Faculty of Earth Sciences and Engineering United States  
**Student Exchange Program – Malaysia Representatives (Distinction).** Sept 2015- Dec 2015  
Subjects: Petroleum Economics, Reservoir Characterization, Technical Communication, General Psychology

### Work Experience

**Inception42 | <https://inception42.ai/>** Abu Dhabi, UAE  
Applied Scientist April 2025-Sept 2026

- **Led Inception's efforts to develop an Agentic Reasoning Multimodal LLM under ADNOC's \$360 million EnergyAI initiative**, integrating well logs and core data in collaboration with MBZUAI researchers to advance petrophysics-domain agentic AI capabilities for the energy sector.
- **Led the development of ADNOC Well Log Bench**, the **first LLM evaluation benchmark in formation evaluation**, for ADNOC carbonate reservoirs.
- **Led post-training of ADNOC Well Log-LLM**, a **domain-specialized long context reasoning model**, using a structured **post-training** pipeline of **SFT → DPO → GRPO → GRCO**, achieving accuracy better than larger frontier-scale models such as **Kimi-K2.6, GPT-5, & Qwen3.5** on long context petrophysical reasoning tasks.
- Co-development with MBZUAI in developing **PetroHarness**, a **custom agentic evaluation and execution harness** for petrophysical interpretation workflows, enabling structured reasoning, tool use, context orchestration, and automated analysis across complex well-log tasks.
- Co-development with MBZUAI team in **designing and developing Petrophys**, a comprehensive custom Python library for adaptive petrophysical workflows, supporting scalable well-log processing, feature extraction, formation-level analysis, and reusable domain-specific interpretation pipelines.
- Devised **scalable long-context solutions** using dynamic feature selection, parallel processing, and geological chunking strategy, enabling **efficient multi-well and cross-formation analysis** over extensive subsurface datasets.
- Integrated training-free optimization methods, including **Group Relative Context Optimization**, to improve reasoning accuracy in a compute-efficient manner without additional model training for ADNOC well logs .

**SLB KL Innovation Factori | [www.slb.com](http://www.slb.com)** Kuala Lumpur, Malaysia  
Senior Domain Data Scientist Nov 2023 – Mar 2025

- **Led for Multilingual-Multimodal Multi Agent** for Exploration & Production, connecting structure and unstructured **Multimodal RAG** to the agentic system for Geoscience subsurface application.

- **Generative AI lead for SLB Asia Domain Data Scientist Community**, in driving multiple project that deliver end-to-end automated GenAI workflows leveraging advance LLM & Multimodal LLM, as well as finetuning domain task specific LLM.
- Leading the domain adapted **post training of Mistral LLM** for operation drilling and geomechanics risk prediction with performance **surpass GPT-4**, achieving performance of **~96% accuracy**.
- Leading the development of integrating deep learning Optical Character Recognition (OCR) with Multimodal LLM for automated reports table extraction, **reducing processing time 70% with accuracy >70%** with Small Language Model.
- Developing **on-prem Handwritten & Multilingual RAG** for document intelligence between Latin, Cyrillic and English languages with **deployed multilingual chatbot**.
- Architected **self-supervised and supervised Hybrid Resnet-Efficient Channel Attention Network**, a high-fidelity pre-stack seismic signal enhancement model for effective swell noise, and linear noise attenuation.
- Pioneering Automated Time-to-Depth Conversion and Velocity Model-Based Uncertainty Analysis by **generating 800 multi-scenarios** for velocity model calibration within 4 hours, and comprehensive data analytics insight for **optimal model selection within 10 minutes**.
- Probabilistic Rock Physics Modelling via ML Optimization, by integrating global sensitivity analysis, derivative free local optimization, and ML clustering for **automated rock physics model calibration and optimal model selection**. The AI SaaS application was deployed via Azure, with integration of Azure Function, SendGrid API, and Postgres SQL connection.
- Real-time distributed fiber optic sensing with edge-enabled streaming analysis, achieving **71% reduction in data size while preserving critical signal** characteristics. Estimated cost saving **~10,000 USD per month**. Engineered **anomaly detection using unsupervised STEGO algorithm** for real-time data.
- Technical Reviewer for Geoscience and Remote Sensing Newsletter, and International Petroleum Technology Conference (IPTC).

Data Analytics Engineer

*Jun 2021 – Nov 2023*

- Technical mentor in developing finetuned pretrained language model tailored for upstream petroleum application, to generate multitask learning model for downstream NLP task. The finetuned pretrained model **surpassing pretrained GPT-2** small and medium model by **10% accuracy**.
- Developed cross domain **Behind Casing Opportunities (BCO) ML model**, achieving F1-score of 74%, 95% for BCO flag, 74% for permeability prediction, and 80% for Net Pay permeability.
- Developed Adaptive Prescriptive Well Performance Intelligence Application by integrating multi-sources data with **> 3 million data points**, for comprehensive field and well level production analysis. Optimizing the big data analytics performance by applying multiprocessing achieving **63% faster processing**, and asynchronous I/O **achieving 87% faster data retrieval**. Adopted as SaaS dashboard for daily production monitoring by engineering team.

**SLB Geosolutions**

*Kuala Lumpur, Malaysia*

Geophysicist

*Mar 2018 – Jun 2021*

- Seismic processing geophysicist, experienced in narrow and wide azimuth surveys at ultra-shallow, deep-water basins for operators e.g. Petronas, Exxon Mobil, ONGC, PTTEP, Santos.
- Expert in seismic deblending, seismic denoise, demultiple, time migration, and velocity picking.
- Developed an automation data extraction from P190 & SEG-D data, **saving > 90% production** time from 2 weeks to 20 minutes.
- Pioneering automated Parameter Analysis and Recommendation for Adaptive Deghosting, together with a Principal Geophysicist, **saving 70% testing time, scalable** for other offshore basin with different acquisition setting, including Offshore Australia, Gulf of Mexico, Gulf of India.

**University of Technology PETRONAS | [www.utp.edu.my](http://www.utp.edu.my)**

*Perak, Malaysia*

Research Assistant – Centre of Seismic Imaging

Assisting Prof Dr. Deva Prasad Ghosh in “Seismic Anisotropy Imaging for Deep Reservoir and Fracture Basement”, book publication on Seismic Interpretation

## **Skills**

- **LLMs & Post-Training:** SFT, DPO, GRPO, long-context reasoning, RAG, agentic tool use, context engineering.
- **Multimodal AI & Computer Vision:** CLIP, Vision Transformers, segmentation, self-supervised learning, contrastive and metric learning, Seismic AI Solution Development

- **Statistics & Optimization:** Sensitivity Analysis, Optimization
- **Engineering & MLOps:** Python, PyTorch, SQL, Docker, FastAPI, MLflow, Weight & Bias, Azure, model deployment and monitoring.
- **Leadership & Delivery:** Team leadership, domain & AI expert collaboration, end-to-end ML implementation.

## Awards & Achievements

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### USA AI Provisional Patent

- Low-Frequency Hybrid Resnet-Efficient Channel Attention (ECA) Network for Pre-stack Linear Noise Attenuation 2025
- AI Approach for Drilling Operation Insights using GenAI LLM 2024
- Generative AI-based Geomechanical Risk Extraction 2024

### Malaysia AI Patent (MY PI2024001424)

- Data Driven Optimization for Rock Physics Modelling Assisted by Machine Learning 2024

**SLB AI & IF Competition Silver Award** - Petronas OSDU Project 2023

**Performed by Schlumberger CEO Award** - Innovation Factori 2022

**Second Runner-Up of EAGE Field Challenge** - Malaysia Representatives from top 10 participants 2017

**AAPG L. Austin Weeks Recipient:** Scholarship recipient for the AAPG Foundation L. Austin Program 2017

**Silver Award in Integrated E&P Opportunity Evaluation** – UTP Final Year E&P Competition 2016

**Winner** – Geo Photography Competition by MOG University Malaya 2015

**First Runner-Up** - Murphy Geo Quiz University Malaya 2015

**First Runner-Up** – Kayak 200m Double Female UTP Sport Carnival 2014

**Petronas Scholarship Recipients:** Top Malaysia Graduates Petronas Scholarship Recipients 2012

## Publications

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- Expert-Reviewed Benchmark and Domain-Specialized LLM for Subsurface Well-Log Interpretation. Authors: Sambal Shikhar, **Lee Ming Xiang**, Zhaksylyk Kabashev, Jean Lahoud, Sandesh Bharadwaj, Sunil Kumar Sahu, Onkar Pandit, Martin Takac, Rao Muhammad Anwer, Federico Castanedo, Hisham Cholakkal. Under review at ICLR 2027 [[Link](#)]
- SPE-236501-MS: Well log LLM Benchmark: ADNOC First Domain-expert-curated Benchmark for LLM Reasoning of Well Logs on Carbonate Reservoir (2026). In Abu Dhabi International Petroleum Exhibition and Conference. Authors: Nader Garges, **Lee Ming Xiang**, L. Barilles, J. Thomas, W. Nasir, S. Bharadwaj, Z. Kabashev, J. Lahoud, O.Pandit , S.K. Sahu, H. Cholakkal, F. Castanedo.
- SPE-236034-MS: PetroHarness: An Open-Source Agentic Harness for Auditable Formation Evaluation (2026). In Abu Dhabi International Petroleum Exhibition and Conference. Authors: S. Shikhar, Z. Kabashev, S. Umurzak, **Lee Ming Xiang**, L. Barillas, S. Bharadwaj, J. Thomas, J. Lahoud, F. Castanedo, H. Cholakkal.
- SPE-229701-MS: Assessing the Capabilities of Large Language Models for Oil and Gas Industry Applications (2025). In Abu Dhabi International Petroleum Exhibition and Conference. Authors: Federico Castanedo, Writa Bjattacharya, Supriyo Ghosh, Martin Takac, S. Lahlou, Z. Iklassov, Markus Schaffrath, N. Reddicharia, R. Mohan, M. Yaslam, **Lee Ming Xiang** [[Link](#)]
- SPE-222023-MS: Retrieving Operation Insights with GenAI LLM: Comparative Analysis and Workflow Enhancement (2024). In Abu Dhabi International Petroleum Exhibition and Conference. Author: **Lee Ming Xiang**, Wang,Z. [[Link](#)]
- Integrating Data Science into Rock Physics Modelling Workflow (2024). In GEO 4.0 Digitalization in Geoscience Symposium, Al Khobar, Saudi Arabia. Authors: **Lee Ming Xiang**, Iakov Kozlov
- SPE-216402-MS: Unsupervised Multitask Learning for Oil and Gas Language Models with Limited Resources (2023). In Abu Dhabi International Petroleum Exhibition and Conference. Authors: Maxime Marlot, Divya Nidhi Srivastava, Fui Kent Wong, **Lee Ming Xiang** [[Link](#)]
- Optimizing Performance in Big Data Handling for Enhanced Data Analytics (2023). In EAGE Workshop on Data Science, Malaysia. Authors: **Lee Ming Xiang**, Atiq Shariff [[Link](#)]
- A survey of Natural Language Processing in Oil and Gas: Opportunities and Challenges (2023). In EAGE Workshop on Data Science, Malaysia. Authors: Maxime Marlot, **Lee Ming Xiang** [[Link](#)]
- Unlocking Value from Text: Visualizing Insights with Natural Language Processing in Unstructured Oil and Gas Reports (2023). In SPE/IATMI Asia Pacific Oil & Gas Conference and Exhibition, Indonesia. Authors: Maxime Marlot, **Lee Ming Xiang**, Asyraf Irfan bin Muhammad Adib, Prasanna Kumar Tellapaneni, Edwin Lawrence [[Link](#)]
- Information Retrieval from Oil and Gas Unstructured Data with Contextualized Framework (2023). In Third EAGE

Digitalization Conference and Exhibition, United Kingdom. Authors: **Lee Ming Xiang**, Maxime Marlot [\[Link\]](#)

- Carbon Dioxide Emission Monitoring based on Prediction of Gas Fuel Rate using Machine Learning (2022). In EAGE Conference on Digital Innovation for a Sustainable Future, Thailand. Authors: **Lee Ming Xiang** [\[Link\]](#)
- Adaptive Deghosting Dashboard (2021). In SLB Technical Coordinators Meeting. Authors: **Lee Ming Xiang**, Artem Sazykin
- Imaging multi-order multiples - Shallow Water Case Study from Southeast Asia (2019). In SEG Kuwait "Seismic Multiples - The Challenges and the Way Forward" Workshop, Kuwait.
- Abstract of Optimum Notch Frequency Recovery using non-CMS approach (2019). In SLB Technical Coordinators Meeting.
- Application of Simultaneous Inversion Characterizing Reservoir Properties in X Field, Sabah Basin (2017). In IOP Conference Series: Earth and Environmental Science, Volume 88, 5th International Conferences on Geological, Geographical, Aerospace and Earth Sciences 2017 (5th AeroEarth 2017), Indonesia. Authors: **Lee Ming Xiang**, Luluan Almann Lubis [\[Link\]](#)